

DP Computer Science: A 1.2.2 Binary Data Storage

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

1. Explain why a **pattern of bits has no meaning** without an agreed **encoding**
2. Explain how **integers**, and how **characters and strings**, are stored in binary, and what determines the space each takes
3. Explain how an **image** is stored in binary using **pixels** and the **RGB model**
4. Explain how **audio** and **video** are stored in binary using **sampling**, and why these files are large
5. Explain the **single principle** connecting all of these: data is stored as fixed-width binary whose meaning is the agreed encoding, which sets the impact on storage and retrieval

Command Term Note: The lesson is held at **Explain**—a detailed account including reasons or causes. The history of Unicode has been cut to a single line; the focus is on the mechanism and storage-cost reasoning.

Key Vocabulary (Tier 3)

Term	Definition
Encoding	The agreed scheme that decides what a pattern of bits means
Character set	A standard that assigns a numeric code to each character (ASCII covers 128; Unicode covers every writing system)
Pixel	The smallest element of an image, holding one colour
RGB model	A colour stored as three values—the strengths of red, green, and blue—each in one byte
Sample rate	How many times per second an audio signal is measured
Bit depth	The number of bits used to store each audio sample

Pedagogical Focus

- **Thinking Skills (ATL):** Students consciously practise **transfer**—they apply one principle (data is a fixed-width binary pattern with an agreed meaning) across five different data types, and state where the storage cost comes from in each.
- **Conceptual Understanding (ATT):** The lesson moves from a **concrete case to an abstract principle** rather than listing five data types side by side. Start from one byte that could be a number, a letter, or a colour value, then build each data type as another instance of that idea.

Name the skill to students: *"Today you are applying one principle across five data types—the transfer is the point."*

EAL Scaffold

Support Type	Description
Sentence Frame	"A [data type] is stored in binary as _____. The space it takes is determined by _____ because _____."
Concept Map	Partially completed with branches drawn and principle written in

2. Lesson Sequence (One-Hour Block)

Part 1: Do Now (Retrieval) — ~10%

Activity: Three questions recalling prior learning:

Question	Source
1. Convert the denary number 65 to 8-bit binary	[A1.2.1]
2. How many different values can 8 bits represent? Show the calculation	[A1.2.1]
3. What does "non-volatile" mean, and why does secondary storage need to be non-volatile?	[A1.1.7]

Part 2: Main Sequence — ~60%

Phase 1: The Concrete Case (~8 mins)

Display One Byte: 01000001

Interpretation	Value
Integer	65
Character (ASCII)	'A'
Colour channel (RGB)	Red value of 65

"A pattern of bits has no meaning until an **encoding** is agreed. Reading data back means knowing both **where** the bits are and **how** they were encoded."

Phase 2: Progressive Abstraction — Five Data Types (~25 mins)

Data Type 1: Integers

Aspect	Description
Mechanism	Stored directly in binary using place value
Fixed width	8, 16, 32, or 64 bits
Size factor	Width determines range and space
Example	45 ₁₀ = 00101101 ₂ (8 bits)

"An integer is stored in a fixed-width binary pattern. The width determines how large a number can be stored and how much space it takes."

Data Type 2: Characters and Strings

Aspect	Description
Mechanism	Each character assigned a numeric code; code stored in binary

Aspect	Description
ASCII	7-bit; 128 characters (English letters, digits, punctuation)
Unicode	Supports characters from every writing system; UTF-8 uses 1–4 bytes per character
Size factor	Number of characters × bytes per character
Example	"Hi" = 01001000 01101001 (ASCII)

Why Unicode Replaced ASCII:

Standard	Coverage	Bytes per Character
ASCII	128 characters (English only)	1 byte
Unicode (UTF-8)	Every writing system	1–4 bytes

"ASCII covered 128 characters; Unicode extends this to every script. UTF-8 keeps the ASCII codes for compatibility."

Data Type 3: Images

Aspect	Description
Mechanism	Stored as a grid of pixels
RGB model	Each pixel = three values: Red, Green, Blue (each 1 byte = 8 bits)
Colour depth	24-bit colour = 3 bytes per pixel = 16.7 million colours
Size factor	Width × height × bytes per pixel
Example	1000 × 1000 × 3 bytes = 3,000,000 bytes

Visualising RGB:

text

Pixel: (R=255, G=0, B=0) = RED

Pixel: (R=0, G=255, B=0) = GREEN

Pixel: (R=0, G=0, B=255) = BLUE

Pixel: (R=255, G=255, B=0) = YELLOW

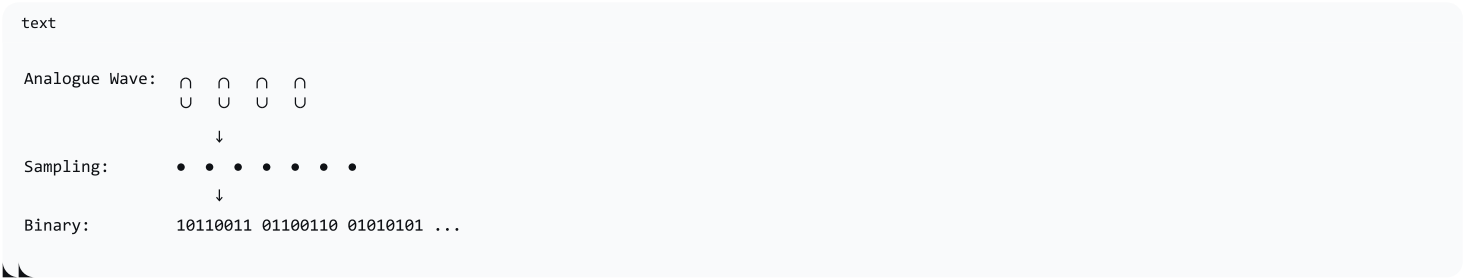
"Each pixel is stored as three binary values. The more pixels (resolution), the more space."

Data Type 4: Audio

Aspect	Description
Mechanism	Analogue sound wave is sampled at regular intervals
Sample rate	How many times per second the signal is measured
Bit depth	Number of bits per sample (quality)

Aspect	Description
Size factor	Sample rate × bit depth × duration
Example	44.1 kHz × 16 bits × 3 minutes = ~15 MB

Visualising Sampling:



"Audio is stored by measuring the sound wave many times per second. More samples (sample rate) and more bits per sample (bit depth) mean larger files."

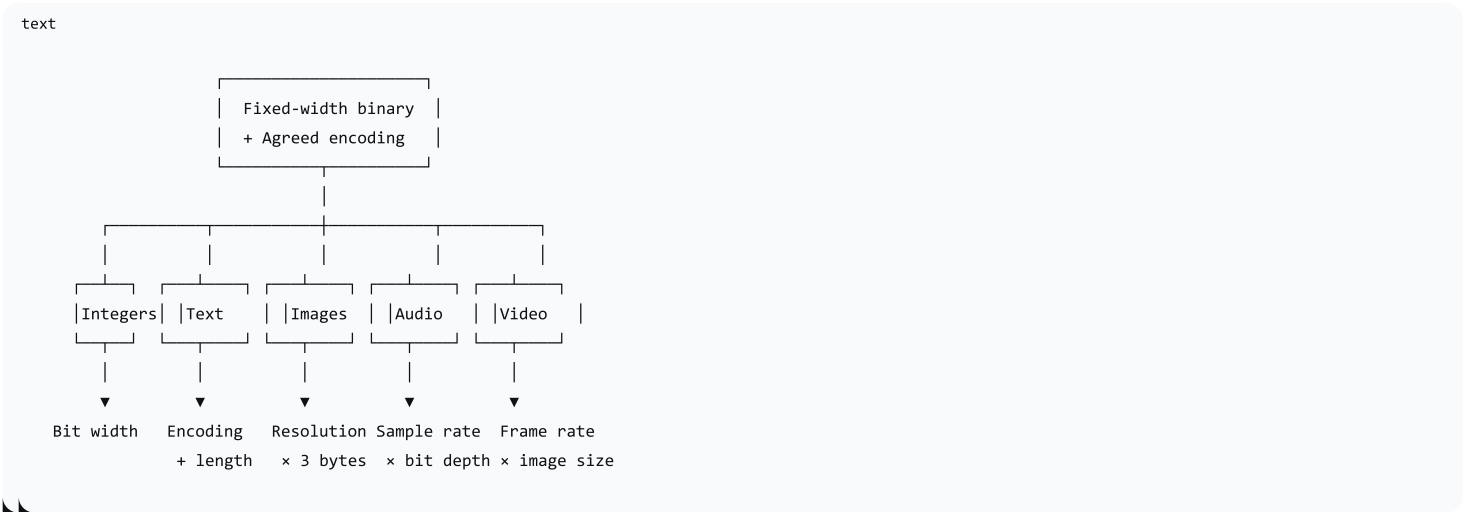
Data Type 5: Video

Aspect	Description
Mechanism	Sequence of frames (individual images)
Frame rate	Frames per second (e.g., 24, 30, 60 fps)
Size factor	Frames per second × image size × duration
Example	30 fps × 1920×1080 × 3 bytes × 60 seconds = ~10 GB (uncompressed)

"Video is a sequence of images (frames). One second holds many full images, which is why files become enormous, and why compression is essential."

Phase 3: Linking Task — Concept Map (~7 mins)

Task: Students build a concept map connecting all five data types to the single principle and to the impact on storage and retrieval.



Part 3: Deliberate Practice — ~20%

Activity: Students write explain-how answers at DP level.

Question 1:

"Explain how a colour image is stored in binary, referring to pixels and the RGB model."

Question 2:

"Explain why one minute of uncompressed audio or video is large."

Sentence Frame:

"[Data type] is stored in binary as _____. The space it takes is determined by _____ because _____. This means that [impact on storage/retrieval]."

Part 4: Exit Check — ~10%

One Explain-How Question:

"Explain how the string 'Hi' is stored in binary, and state what would change its size."

Expected Answer:

"Each character is stored using a binary code. In ASCII, 'H' is 72 (01001000) and 'i' is 105 (01101001). The size is determined by the encoding and the number of characters; using UTF-8 instead of ASCII, or adding more characters, would increase its size."

Readable in minutes and used to plan the next lesson.

3. Summary: Data Types at a Glance

Data Type	Storage Mechanism	Size Factor	Example
Integer	Direct binary (fixed width)	Bit width (8, 16, 32, 64)	45 ₁₀ → 00101101 ₂
Text	Character codes (ASCII/Unicode)	Encoding × number of characters	"Hi" → 01001000 01101001
Image	Pixel grid + RGB values	Width × height × 3 bytes	1000×1000 → 3 MB
Audio	Sampling + quantisation	Sample rate × bit depth × time	44.1kHz × 16-bit → ~15 MB/min
Video	Sequence of frames	Frame rate × image size × time	30 fps × 1080p → ~10 GB/min

4. The Single Principle

- All data stored in a computer is represented using binary (0s and 1s).
- The meaning of any binary pattern comes from the agreed encoding.
- The storage cost depends on the fixed width of the representation and the number of items.

Principle	Application
Fixed-width binary	Integers use 8, 16, 32, or 64 bits; characters use 1–4 bytes
Agreed encoding	ASCII, Unicode, RGB, sample rates are all encodings
Storage cost	More width, more characters, more pixels, more samples, or more frames = larger files

5. Differentiation & Extension

Support (Scaffolds That Can Be Removed)

Support Strategy	Description
Partially Completed Concept Map	Five branches drawn, principle written in; students add mechanism and size factor
ASCII Codes Provided	For the worked string example
Sentence Frames	For written explanations

Challenge (Deeper or Wider)

Challenge Strategy	Description
Calculate Uncompressed Size	Calculate the uncompressed size of a given image or audio clip from its parameters
Explain 24-Bit Colour	Explain, using the RGB model, why 24-bit colour gives 16,777,216 possible colours
Global Context Link	Discuss why a single universal character set such as Unicode matters for text in every writing system

6. Example Hinge Questions

Question	Purpose
"Why is it necessary to convert information into discrete units for computers to process it?"	Checks understanding of digital data
"Can you give an example of how binary code might be used in a real-world application, besides storing numbers?"	Encourages thinking beyond simple number representation
"How does the binary representation of an image differ from that of a piece of text?"	Encourages comparison of different data types
"Why is a 24-bit colour image larger than an 8-bit grayscale image?"	Checks understanding of bit depth and storage cost
"What determines the size of an audio file?"	Checks understanding of sample rate and bit depth
"Why does one second of video require so much storage space compared to one second of audio?"	Checks understanding of video storage mechanism

7. Learning Objectives (Summary)

Knowledge and Understanding

- Explain the concept of binary encoding and how it uses only two states (0 and 1) to represent all data
- Describe the relationship between binary code and the on/off states of electrical signals in a computer

- Explain the impact of binary encoding on data storage, retrieval, and processing

Data Storage Mechanisms

- **Integers:** Positive integers represented directly using binary code
- **Characters/Text:** Encoded using ASCII and Unicode
- **Images:** Grid of pixels; each pixel's colour encoded as binary value(s)
- **Audio:** Digitised by sampling analogue signals
- **Video:** Sequence of frames (images) plus audio

ATL Skills

Skill	Description
Research	Investigate different data encoding schemes and their applications
Thinking	Apply critical thinking to analyse how different data types are represented and stored
Communication	Explain binary encoding and its impact on data storage and retrieval

8. Resources

- **Presentation** (Slide Deck)
- **InThinking eBook** (A1.2.1–A1.2.2 with quiz)
- **Worksheet:** Concept map + explain-how questions
- **Worksheet Answers**

Author's Reflection

"The organising idea for the lesson is that a pattern of bits means nothing until we agree an encoding (data becoming information). The lesson opens with a single byte that is a number, a letter, and a colour value at once, and only the agreed encoding decides which. Build the five data types as instances of that one idea rather than five things to memorise, because the transfer is the point: once students see that everything is a fixed-width pattern with an agreed meaning, integers, text, images, audio, and video stop being separate topics and become the same topic asked five ways.

The old deck spent several slides on where Unicode came from, and at Explain level the statement wants the mechanism—a character is given a code and the code is stored as bits, and the reason one file is larger than another. I have cut the history to a single line and put that time into the storage-cost reasoning instead. Video was missing and the statement names it, so I have added it as the cleanest case of the principle: a frame is just an image, and a second of video is many frames, which is why the numbers become enormous and why compression, met earlier in A1.1, exists."